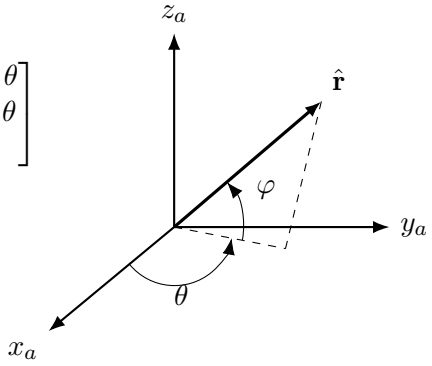
Quantitative measure of the force model

步驟 1

1.1 單位方向 $\hat{\mathbf{r}}(\varphi,\theta)$

以致動座標系的基底 $\{\hat{\mathbf{x}}_a,\ \hat{\mathbf{y}}_a,\ \hat{\mathbf{z}}_a\}$ 描述任意單位方向 $\hat{\mathbf{r}}$：

$$\hat{\mathbf{r}}(\varphi,\theta)=\cos\varphi\cos\theta\,\hat{\mathbf{x}}_a+\cos\varphi\sin\theta\,\hat{\mathbf{y}}_a+\sin\varphi\,\hat{\mathbf{z}}_a=\begin{bmatrix}\cos\varphi\cos\theta\\ \cos\varphi\sin\theta\\ \sin\varphi\end{bmatrix}$$



$$\begin{cases}\varphi\in[-90^\circ,\ 90^\circ] & \text{仰角（自 } x_a\text{–}y_a \text{ 平面起算）}\\ \theta\in[0^\circ,\ 360^\circ) & \text{方位角（自 } +x_a \text{ 軸起算，繞 } z_a\text{）}\end{cases}$$

$$\|\hat{\mathbf{r}}\|=1$$

1.2 電流 $\mathbf{I}$

位置 $\mathbf{p}_i$、電流 $\mathbf{I}$ 產生的力向量

$$\mathbf{f}_i=\begin{bmatrix}f_{i1}\\ f_{i2}\\ f_{i3}\end{bmatrix},\qquad f_{ik}=\mathbf{I}^{\mathsf{T}}\,{}^{F}\hat{\mathbf{H}}_I^{\mathsf{T}}\,\mathbf{L}_k\Big(\mathbf{p}_i/\hat{\ell},\,\hat{\mathbf{e}}\Big)\,{}^{F}\hat{\mathbf{H}}_I\,\mathbf{I},\qquad k=1,2,3$$

令

$$\mathbf{I}=\sqrt{\|\mathbf{f}_i\|}\,\hat{\mathbf{I}},\qquad \mathbf{f}(\mathbf{p}_i,\mathbf{I})=\|\mathbf{f}_i\|\,\hat{\mathbf{r}}$$

代入力模型：

$$\mathbf{f}_i=\begin{bmatrix}\Big(\sqrt{\|\mathbf{f}_i\|}\,\hat{\mathbf{I}}\Big)^{\mathsf{T}}\,{}^{F}\hat{\mathbf{H}}_I^{\mathsf{T}}\,\mathbf{L}_1\Big(\mathbf{p}_i/\hat{\ell},\hat{\mathbf{e}}\Big)\,{}^{F}\hat{\mathbf{H}}_I\,\Big(\sqrt{\|\mathbf{f}_i\|}\,\hat{\mathbf{I}}\Big)\\ \Big(\sqrt{\|\mathbf{f}_i\|}\,\hat{\mathbf{I}}\Big)^{\mathsf{T}}\,{}^{F}\hat{\mathbf{H}}_I^{\mathsf{T}}\,\mathbf{L}_2\Big(\mathbf{p}_i/\hat{\ell},\hat{\mathbf{e}}\Big)\,{}^{F}\hat{\mathbf{H}}_I\,\Big(\sqrt{\|\mathbf{f}_i\|}\,\hat{\mathbf{I}}\Big)\\ \Big(\sqrt{\|\mathbf{f}_i\|}\,\hat{\mathbf{I}}\Big)^{\mathsf{T}}\,{}^{F}\hat{\mathbf{H}}_I^{\mathsf{T}}\,\mathbf{L}_3\Big(\mathbf{p}_i/\hat{\ell},\hat{\mathbf{e}}\Big)\,{}^{F}\hat{\mathbf{H}}_I\,\Big(\sqrt{\|\mathbf{f}_i\|}\,\hat{\mathbf{I}}\Big)\end{bmatrix}$$

$$=\|\mathbf{f}_i\|\begin{bmatrix}\hat{\mathbf{I}}^{\mathsf{T}}\,{}^{F}\hat{\mathbf{H}}_I^{\mathsf{T}}\,\mathbf{L}_1\Big(\mathbf{p}_i/\hat{\ell},\hat{\mathbf{e}}\Big)\,{}^{F}\hat{\mathbf{H}}_I\,\hat{\mathbf{I}}\\ \hat{\mathbf{I}}^{\mathsf{T}}\,{}^{F}\hat{\mathbf{H}}_I^{\mathsf{T}}\,\mathbf{L}_2\Big(\mathbf{p}_i/\hat{\ell},\hat{\mathbf{e}}\Big)\,{}^{F}\hat{\mathbf{H}}_I\,\hat{\mathbf{I}}\\ \hat{\mathbf{I}}^{\mathsf{T}}\,{}^{F}\hat{\mathbf{H}}_I^{\mathsf{T}}\,\mathbf{L}_3\Big(\mathbf{p}_i/\hat{\ell},\hat{\mathbf{e}}\Big)\,{}^{F}\hat{\mathbf{H}}_I\,\hat{\mathbf{I}}\end{bmatrix}$$

$$=\|\mathbf{f}_i\|\,\hat{\mathbf{f}}\Big(\mathbf{p}_i,\hat{\mathbf{I}}\Big)$$

比較得

$$\hat{\mathbf{f}}\Big(\mathbf{p}_i,\hat{\mathbf{I}}\Big)=\hat{\mathbf{r}}$$

$\hat{\mathbf{f}}(\mathbf{p}_i,\hat{\mathbf{I}})$ 稱為單位力量，所對應的電流 $\hat{\mathbf{I}}$ 稱為單位電流。

步驟 2

期望力 $\mathbf{f}_d$ 拆成大小與方向：

$$\mathbf{f}_d=\|\mathbf{f}_d\|\,\hat{\mathbf{r}}(\varphi,\theta)$$

$$\mathbf{I}_{\text{opt}} = \arg\min_{\mathbf{I}} \ \|\mathbf{I}\|^2 \qquad \text{s.t.} \quad \mathbf{f}(\mathbf{p}_i, \mathbf{I}) = \mathbf{f}_d(\mathbf{p}_i, \mathbf{I}_{\text{opt}})$$

以 $\mathbf{I} = \sqrt{\|\mathbf{f}_i\|}\,\hat{\mathbf{I}}$ 替換 $\mathbf{I}$：

$$\|\mathbf{I}\|^2 = \|\mathbf{f}_i\|\,\|\hat{\mathbf{I}}\|^2, \qquad \mathbf{f}(\mathbf{p}_i, \mathbf{I}) = \|\mathbf{f}_i\|\,\hat{\mathbf{f}}\Big(\mathbf{p}_i, \hat{\mathbf{I}}\Big) = \mathbf{f}_d$$

限制式給出 $\|\mathbf{f}_i\| = \|\mathbf{f}_d\|$、$\hat{\mathbf{f}}(\mathbf{p}_i, \hat{\mathbf{I}}) = \mathbf{f}_d/\|\mathbf{f}_d\| = \hat{\mathbf{r}}$，$\|\mathbf{f}_d\|$ 為常數。
換成最小範數單位電流問題：

$$\hat{\mathbf{I}}_{\text{opt}} = \arg\min_{\hat{\mathbf{I}}} \ \|\hat{\mathbf{I}}\|^2 \qquad \text{s.t.} \quad \hat{\mathbf{f}}\Big(\mathbf{p}_i, \hat{\mathbf{I}}\Big) = \hat{\mathbf{r}}$$

原問題的解：

$$\mathbf{I}_{\text{opt}} = \sqrt{\|\mathbf{f}_d\|}\,\hat{\mathbf{I}}_{\text{opt}}$$

步驟 3

固定位置 $\mathbf{p}_i$，以 Lagrange multipliers 求解任意方向 $\hat{\mathbf{r}}$ 的 $\hat{\mathbf{I}}_{\text{opt}}(\mathbf{p}_i, \varphi, \theta)$。令

$$\mathbf{Q}_k = {}^{F}\hat{\mathbf{H}}_I^{\mathsf{T}}\,\mathbf{L}_k\Big(\mathbf{p}_i/\hat{\ell}, \hat{\mathbf{e}}\Big)\,{}^{F}\hat{\mathbf{H}}_I, \qquad k = 1, 2, 3$$

限制式 $\hat{\mathbf{f}}(\mathbf{p}_i, \hat{\mathbf{I}}) = \hat{\mathbf{r}}$ 的第 k 個分量為 $\hat{\mathbf{I}}^{\mathsf{T}}\mathbf{Q}_k\hat{\mathbf{I}} = r_k$，Lagrange multipliers $\boldsymbol{\lambda} = \begin{bmatrix}\lambda_1 & \lambda_2 & \lambda_3\end{bmatrix}^{\mathsf{T}}$：

$$\mathcal{L}\Big(\hat{\mathbf{I}}, \boldsymbol{\lambda}\Big) = \|\hat{\mathbf{I}}\|^2 - \sum_{k=1}^{3} \lambda_k \left(\hat{\mathbf{I}}^{\mathsf{T}}\mathbf{Q}_k\hat{\mathbf{I}} - r_k\right)$$

步驟 4

由 Lagrange multipliers 得到 $\hat{\mathbf{I}}_{\text{opt}}$，令正規化電流

$$\bar{\mathbf{I}} = \frac{\hat{\mathbf{I}}_{\text{opt}}}{\left\|\hat{\mathbf{I}}_{\text{opt}}\right\|}, \qquad \|\bar{\mathbf{I}}\| = 1$$

以大小為 $\|\mathbf{I}\|$（A）的電流 $\mathbf{I} = \|\mathbf{I}\|\,\bar{\mathbf{I}}$ 代入力模型：

$$\mathbf{f}(\mathbf{p}_i, \mathbf{I}(\varphi, \theta)) = \|\mathbf{I}\|^2 \begin{bmatrix}\bar{\mathbf{I}}^{\mathsf{T}}\mathbf{Q}_1\bar{\mathbf{I}} \\ \bar{\mathbf{I}}^{\mathsf{T}}\mathbf{Q}_2\bar{\mathbf{I}} \\ \bar{\mathbf{I}}^{\mathsf{T}}\mathbf{Q}_3\bar{\mathbf{I}}\end{bmatrix}$$

$$= \frac{\|\mathbf{I}\|^2}{\left\|\hat{\mathbf{I}}_{\text{opt}}\right\|^2} \begin{bmatrix}\hat{\mathbf{I}}_{\text{opt}}^{\mathsf{T}}\mathbf{Q}_1\hat{\mathbf{I}}_{\text{opt}} \\ \hat{\mathbf{I}}_{\text{opt}}^{\mathsf{T}}\mathbf{Q}_2\hat{\mathbf{I}}_{\text{opt}} \\ \hat{\mathbf{I}}_{\text{opt}}^{\mathsf{T}}\mathbf{Q}_3\hat{\mathbf{I}}_{\text{opt}}\end{bmatrix} = \frac{\|\mathbf{I}\|^2}{\left\|\hat{\mathbf{I}}_{\text{opt}}\right\|^2}\,\hat{\mathbf{f}}\Big(\mathbf{p}_i, \hat{\mathbf{I}}_{\text{opt}}\Big) = \frac{\|\mathbf{I}\|^2}{\left\|\hat{\mathbf{I}}_{\text{opt}}\right\|^2}\,\hat{\mathbf{r}}(\varphi, \theta)$$

定義

$$\rho(\mathbf{p}_i, \hat{\mathbf{r}}) = \frac{1}{\left\|\hat{\mathbf{I}}_{\text{opt}}\right\|^2} \qquad (\text{pN/A}^2)$$

代回力量：

$$\mathbf{f}(\mathbf{p}_i, \mathbf{I}(\varphi, \theta)) = \|\mathbf{I}\|^2\,\rho(\mathbf{p}_i, \hat{\mathbf{r}})\,\hat{\mathbf{r}}(\varphi, \theta)$$

$\rho(\mathbf{p}_i, \hat{\mathbf{r}})$ 為在位置 $\mathbf{p}_i$、沿方向 $\hat{\mathbf{r}}$ 所能達到之力增益的最大值（pN/A^2）。

步驟 5

$$\mathcal{C}_f(\mathbf{p}_i) = \left(\frac{1}{4\pi} \int_{-\pi/2}^{\pi/2} \int_{0}^{2\pi} \rho^3(\mathbf{p}_i, \hat{\mathbf{r}}(\varphi, \theta)) \cos\varphi \; d\theta \, d\varphi \right)^{1/3} \qquad (\mathrm{pN/A^2})$$

$$\kappa_f(\mathbf{p}_i) = \frac{\min\limits_{\hat{\mathbf{r}}} \; \rho(\mathbf{p}_i, \hat{\mathbf{r}})}{\max\limits_{\hat{\mathbf{r}}} \; \rho(\mathbf{p}_i, \hat{\mathbf{r}})}$$